

Soil node PCB technical specification

AmudarIO | Board 1 | Version 1.0 | 4 September 2026 | Pinout C6-P1

Design an autonomous PCB for one external 5 V RS-485 soil probe. The node measures soil temperature, volumetric water content, electrical conductivity and battery voltage, and sends the results to ChirpStack using LoRaWAN Class A.

1 Hardware and architecture

Block	Implementation
MCU	ESP32-C6-MINI-1-H4, 4 MB flash
LoRa	SX1262-based DX-LR30-900M22S; external antenna through SMA
Battery	Removable protected 18650 Li-ion cell, 1S, 3.6/3.7 V nominal, 4.2 V charge
Holder	MPD BK-18650-PC2 with retainer; holder included in PCBA, cell supplied separately
Solar charging	LTC4079 with hardware NTC in physical contact with the cell
Always-on 3.3 V	TPS63900 buck-boost for ESP32 and LoRa
Switched 5 V probe rail	TPS61023 with EN and true output disconnect
RS-485	THVD1406 automatic direction; power gated by TPS22917
Battery measurement	Internal ADC; divider with TPS22917 high-side switch
Service	External USB-to-TTL on UART0; BOOT, RESET and SERVICE WAKE

A nominal 6 V solar panel feeds the charger. The protected battery feeds the 3.3 V and 5 V converters through a mechanical SYSTEM switch. This switch disconnects all system loads; the charger and NTC remain connected to the panel and battery.

Do not fit GPS, an external RTC, ADS1115, a fuel gauge, continuously lit LEDs or an onboard USB-UART bridge. Disable Wi-Fi, BLE and 802.15.4 in deployed firmware. LiFePO₄ is not an interchangeable battery option.

2 Operating cycle

Timer wake; safe initialization; enable probe and RS-485; wait for startup; read Modbus with bounded retries; disable the probe; measure battery voltage; send uplink; open RX1/RX2; process a command and acknowledge immediately; shut down LoRa; enter deep sleep. The server can change the interval, which must be retained. Do not execute a repeated command with the same ID.

3 Complete ESP32 C6 pinout

GPIO is the firmware identifier. Pad is the physical ESP32-C6-MINI-1-H4 module pad, not a bare-chip pin or DevKit header position. A = ADC input, I = input, O = output. All logic signals use 3.3 V levels.

GPIO	Pad	I/O	Net	Function
0	12	A	VBAT_ADC	Battery voltage, ADC1_CH0
1	13	O	VBAT_DIV_EN	Battery divider high-side switch enable
2	5	O	SENSOR_DOMAIN_EN	Shared enable for 5 V boost and RS-485 power
3	6	O	LORA_RXEN	LoRa RF switch receive control
4	9	-	RESERVED_STRAP_MTMS	Reserved SDIO strap / MTMS
5	10	-	RESERVED_STRAP_MTDI	Reserved SDIO strap / MTDI
6	15	I	SERVICE_WAKE_N	Wake button, connects to GND
7	16	O	LORA_TXEN	LoRa RF switch transmit control
8	22	-	RESERVED_BOOT_STRAP	Reserved boot strap
9	23	I	BOOT_N	BOOT button, programming and recovery only
12	17	O	RS485_TX	UART1 TX to automatic-direction RS-485
13	18	I	RS485_RX	UART1 RX from automatic-direction RS-485
14	19	O	LORA_RESET_N	LoRa reset, active LOW
15	20	-	RESERVED_JTAG_STRAP	Reserved JTAG source-selection strap
16	31	O	UART0_TX	UART0 TX, logs and external USB-to-TTL
17	30	I	UART0_RX	UART0 RX, programming and Serial commands
18	24	O	LORA_SCK	LoRa SPI clock
19	25	I	LORA_MISO	SPI data from LoRa to MCU
20	26	O	LORA_MOSI	SPI data from MCU to LoRa
21	27	O	LORA_NSS	LoRa SPI chip select, active LOW
22	28	I	LORA_DIO1	LoRa interrupt
23	29	I	LORA_BUSY	LoRa busy status

Power: pad 3 = 3.3 V; pad 8 = EN/CHIP_PU. GND: pads 1, 2, 11, 14, 36-53. NC: pads 4, 7, 21, 32-35. GPIO10/11 are not exposed. No general-purpose GPIOs remain unallocated. Fit an EN RC network and supervisor suitable for slow and interrupted power ramps.

GPIO1 enables only the battery-divider switch. GPIO2 enables the 5 V boost and RS-485 power switch together. GPIOs must not carry load current or connect directly to the battery, 5 V or A/B lines.

4 GPIO restrictions and safe states

- GPIO4/5/8/9/15 are strapping pins. Do not load GPIO4/5. Pull GPIO8/9 to 3.3 V through 10 kOhm and GPIO15 to GND through 10 kOhm. UART download requires GPIO8=1 and GPIO9=0 at reset; normal boot requires GPIO9=1.
- GPIO6/7 reuse alternate JTAG functions. Do not enable external pad JTAG on GPIO4-7. Check the factory eFuse configuration; do not burn eFuses to accommodate the pinout.
- GPIO12/13 default to USB D-/D+. Disable USB functions and pulls, then assign UART1 before enabling RS-485. BOTH UART paths need powered-off isolation with specified Ioff. A series resistor alone is insufficient.
- Reserve GPIO16/17 for UART0. ROM TX logs must not control loads. GPIO18-23 are valid for LoRa, but require explicit SPI routing and consideration of startup SDIO pulls.

Signals	Reset and startup	Deep sleep
GPIO1/2	LOW, external PD about 100 kOhm	LOW and held; divider and sensor domain OFF
GPIO3/7 RXEN/TXEN	LOW, PD about 100 kOhm	LOW and held
GPIO14/21 RESET/NSS	HIGH, PU 47-100 kOhm	HIGH after radio shutdown
GPIO18/20 SCK/MOSI	Keep NSS HIGH	LOW and held
GPIO6 WAKE	PU about 100 kOhm; button to GND	Wake input; HIGH when idle
ADC and other inputs	No overvoltage or back-power	Disable unused input buffers

External circuitry must establish these states; firmware alone is insufficient. Before sleep, disconnect UART paths, set GPIO2/1 and RXEN/TXEN LOW, and configure hold. On wake, establish safe levels before releasing hold. Confirm pull values against leakage and noise margins.

5 LoRa connections

Signal	DX-LR30 pin	ESP32 GPIO
RXEN	6	3
TXEN	7	7
DIO1	13	22
BUSY	14	23
RESET	15	14
MISO	16	19
MOSI	17	20
SCK	18	18
NSS	19	21

DX-LR30 supply: 3.3 V, pin 9 in the project reference. Leave DIO2/DIO3 unconnected. Verify the purchased module marking and revision drawing before assigning its footprint. LoRa module pin numbers are not ESP32 GPIO numbers.

6 Power charging and battery measurement

Configure LTC4079 for 4.2 V and fixed input-voltage regulation based on the selected panel Vmp. It is a linear charger with Vmp regulation, not dynamic MPPT. Charge current must not exceed 250 mA or the cell, panel and thermal limits. Verify weak-light startup and charge termination with the node operating. Hardware NTC qualification must inhibit charging outside the cell temperature limits and on NTC faults.

If winter energy is insufficient, LTC4121-4.2 with fractional-Voc MPPT is an approval-required alternative after recalculating sleep current. Cell overcharge, over-discharge, overload and short-circuit protection are mandatory independently of the charger. Provide reverse-insertion protection, a cell retainer and an NTC contacting the cell sidewall.

Validate TPS63900 at minimum battery voltage and ESP32/LoRa peak load. The TPS61023 stage must disconnect and discharge its 5 V output; protect against shorts and reverse back-power. All disabled interfaces must remain high impedance.

Switch the battery divider on its high side. Starting values: 499 + 499 kOhm upper leg, 249 kOhm lower leg, 100 nF from ADC to GND. GPIO1 enables the switch; GPIO0 reads ADC1_CH0. Wait at least five RC time constants, discard the first sample, average readings and disable the switch. Confirm values for leakage, ADC loading and temperature; perform two-point calibration. Report voltage and normal/low/critical states, not an accurate state-of-charge percentage.

7 Connectors and RS485

Probe connector: 1 = +5V_SOIL_SW; 2 = GND/reference; 3 = RS485_A; 4 = RS485_B. Panel connector: PV+ and PV-. Use keyed, retained connectors with polarity labels. Select the connector family for the enclosure and cable.

THVD1406 direction is automatic; no MCU DE/RE is allocated. At the connector, fit an SM712-class TVS and footprints for surge-rated series resistors, optional 120 ohm termination and bias. Select values and termination population for the cable; power bias from the switched domain. Validate 4800/9600 baud, bus release timing and Modbus with the actual probe.

8 Programming and Serial Monitor

Pin	Board signal	Adapter connection
1	GND	GND
2	3V3_REF	High-impedance reference input only; otherwise leave open
3	UART0_TX / GPIO16	Adapter RX
4	UART0_RX / GPIO17	Adapter TX
5	DTR	Through the two-transistor auto-reset circuit
6	RTS	Through the two-transistor auto-reset circuit

Use 3.3 V logic only. Power the board from its own source; do not connect the adapter 3.3 V, 5 V or VBUS power outputs. BOOT: GPIO9; RESET: EN, module pad 8. Route DTR/RTS through the standard Espressif circuit, never directly to BOOT/EN. With a TX/RX/GND-only adapter, use the BOOT and RESET buttons. The same UART0 supports Serial Monitor; verify DTR/RTS behavior when opening the port.

9 PCB and power consumption requirements

Baseline: two-layer FR-4 with a nearly continuous ground plane. Four layers require a routing or EMC justification. ANT-to-SMA must be a short 50 ohm controlled-impedance trace calculated by the fabricator for the actual stackup. This is trace geometry, not a 50 ohm resistor. Keep power loops and inductors away from RF/ADC, respect the ESP32 antenna keepout, and locate protection at the connectors.

Mode	Requirement
Normal timer deep sleep	Complete assembled-board current no greater than 25 uA
Inactive state and error backoff	Below 1 mA; probe, RS-485 and divider OFF
SYSTEM OFF	System loads unpowered; charging qualified by hardware NTC
Sensing, boot, LoRa TX/RX, programming, charging	Active modes; sleep-current limits do not apply

Measure at the battery input with SYSTEM ON, panel and programmer disconnected, LEDs off, LoRa shut down and probe power removed. Use 3.70 V as the reference point; repeat at voltage and temperature limits. Also test with the unpowered probe connected. Include the cell protector, charger, supervisor, pulls and leakage in the budget. Provide a removable current-measurement link. Bound join/Modbus retries, then enter sleep.

10 Verification and contractor deliverables

- Verify power-up, BOOT/RESET, programming, watchdog and brownout without unintended probe power. Verify no back-power through UART or A/B.
- Test charging, NTC faults and temperature, 3.3 V under RF peaks, 5 V during probe startup and short circuit, ADC, Modbus, LoRaWAN join/uplink/RX1/RX2/ACK, sleep and wake.
- Deliver native EDA files and libraries, schematic PDF, BOM with exact manufacturer part numbers, Gerber/NC drill, assembly drawings and pick-and-place, STEP, ERC/DRC, power/RF/autonomy calculations, test firmware and measurement reports.

11 Design inputs and component selection

Before fabrication, record the protected 18650 model and dimensions, NTC and temperature limits; panel Voc/Vmp/Isc and power; probe model, Modbus map, current and startup time; DX-LR30 revision and regional plan; enclosure, outline, connectors and cables; winter autonomy, intervals, battery thresholds, rail/ADC tolerances and EMC/ESD levels. The contractor shall calculate component values and obtain customer approval.

Component references: Espressif ESP32-C6-MINI-1 datasheet and Hardware Design Guidelines; Analog Devices LTC4079; Texas Instruments TPS63900, TPS61023, TPS22917 and THVD1406. The pin-map identifier for this specification is C6-P1.